Note on A course in Arithmetic by J.-P. Serre

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Theorem 1.1
The finite subgroup of a field's multiplicative group is cyclic.

This can be easily proven using the structure theorem of finite abelian groups and the fundamental theorem of Algebra.

Definition 1.2. For any prime p or $p = \infty$, we define the

$$\|x\|_p = \begin{cases} p^{-v_p(x)} & p \text{ is a prime} \\ \|x\| & p = \infty \end{cases}$$

Theorem 1.3
For any rational number r, $\prod_p \|r\|_p = 1$

This is trivial from the definition of v_p.

We can in general define a norm on a number field K as follows:

Definition 1.4. $\|\cdot\| : K \to \mathbb{R}_{\geq 0}$ is a norm if

1. $\|x\| = 0$ iff $x = 0$
2. $\|xy\| = \|x\| \|y\|$
3. $\|x + y\| \leq \|x\| + \|y\|$

Corollary 1.5
$\|1\| = 1$ and $\|-1\| = 1$

Definition 1.6. Norm N_1 and N_2 are equivalent if there exists a real number c such that $N_2(x) = N_1(x)^a$ for all x.

Theorem 1.7 (Ostrowski's Theorem)
Any non-trivial norm on $\mathbb{Q}$ is equivalent to either the usual absolute value or to $\|x\|_p$ for some prime p.

Proof. □

Theorem 1.8 (Hensel's Lemma)

Proof. □

Theorem 1.9 (Hasse principle)

Example 1.10

$f(x, y, z) = 5x^2 + 7y^2 - 13z^2$. We want to find a nontrivial $f(x, y, z) = 0$ has non-trivial solutions in $\mathbb{Q}$ using Hensel's Lemma.

Consider this curve in $\mathbb{P}^3$, is there any other solution?

Use parameterization of this curve.

Theorem 1.11

If $n \geq 3$, $x^n + y^n = z^n$ has no non-trivial solutions in $\mathbb{C}[x]$.